ECA0801 - ANALOG AND DIGITAL COMMUNICATION

Exp 2: CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

Aim:

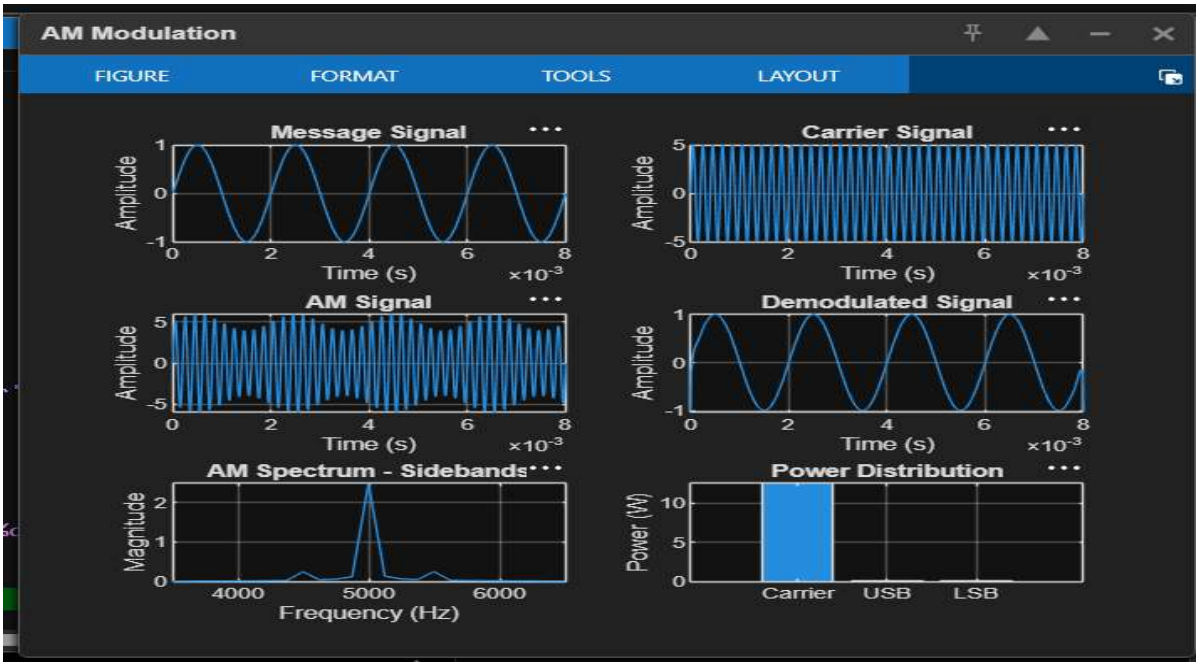
i) To generate an Amplitude Modulated (AM) signal in MATLAB using specified carrier and modulating signal parameters.

ii) To determine and verify the upper sideband (USB) and lower sideband (LSB) frequencies of the AM signal using frequency spectrum analysis in MATLAB.

iii) To calculate the carrier power, sideband power, and total transmitted power of the AM signal for a given modulation index using MATLAB.

iv) To analyze the effect of varying the modulation index on sideband power distribution and overall transmission efficiency through MATLAB simulation.

Apparatus Required: MATLAB Software



Program:

```
clc;
clear;
close all;
% Parameters
Am = 1;
fm = 500;
Ac = 5;
fc = 5000;
fs = 20*fc;
% Time vector
t = 0:1/fs:4/fm;
% Message and carrier signals
m = Am*sin(2*pi*fm*t);
c = Ac*sin(2*pi*fc*t);
% AM modulation
mu = Am/Ac;
am = Ac*(1 + mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);
% Demodulation using envelope detection
dem = abs(hilbert(am));
dem = dem - mean(dem);
% FFT calculation
N = length(am);
f = (-floor(N/2):ceil(N/2)-1)*(fs/N);
AM_FFT = abs(fftshift(fft(am)))/N;
% Sideband frequencies
USB = fc + fm;
LSB = fc - fm;
% Power calculation
Pc = Ac^2/2;
Psb = (mu^2*Pc)/4;
% Plotting
figure('Name','AM Modulation','NumberTitle','off');
subplot(3,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,3);
plot(t,am);
title('AM Signal');
```

```
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,4);
plot(t,dem);
title('Demodulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(3,2,5);
plot(f,AM_FFT);
xlim([fc-3*fm fc+3*fm]);
title('AM Spectrum - Sidebands');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
subplot(3,2,6);
bar([Pc Psb Psb]);
set(gca,'XTick',1:3,'XTickLabel',{'Carrier','USB','LSB'});
title('Power Distribution');
ylabel('Power (W)');
grid on;
% Display results
fprintf('mu = %.2f | USB = %d Hz | LSB = %d Hz | Pc = %.2f W | Psb = %.3f W\n', ...
mu, USB, LSB, Pc, Psb);
```

Output:

